

High abundance of plasma cells secreting transglutaminase 2-specific IgA autoantibodies with limited somatic hypermutation in celiac disease intestinal lesions

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Evidence abstract

Celiac disease is an immune-mediated disorder in which mucosal autoantibodies to the enzyme transglutaminase 2 (TG2) are generated in response to the exogenous antigen gluten in individuals who express human leukocyte antigen HLA-DQ2 or HLA-DQ8 (ref.

3).

We assessed in a comprehensive and nonbiased manner the IgA anti-TG2 response by expression cloning of the antibody repertoire of ex vivo-isolated intestinal antibody-secreting cells (ASCs).

We found that TG2-specific plasma cells are markedly expanded within the duodenal mucosa in individuals with active celiac disease.

TG2-specific antibodies were of high affinity yet showed little adaptation by somatic mutations.

Unlike infection-induced peripheral blood plasmablasts, the TG2-specific ASCs had not recently proliferated and were not short-lived ex vivo.

Altogether, these observations demonstrate that there is a germline repertoire with high affinity for TG2 that may favor massive generation of autoreactive B cells.

TG2-specific antibodies did not block enzymatic activity and served as substrates for TG2-mediated crosslinking when expressed as IgD or IgM but not as IgA1 or IgG1.

This could result in preferential recruitment of plasma cells from naive IgD- and IgM-expressing B cells, thus possibly explaining why the antibody response to TG2 bears signs of a primary immune response despite the disease chronicity.

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Evidence checklist

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9. This could result in preferential recruitment of plasma cells from naive IgD- and IgM-expressing B cells, thus possibly explaining why the antibody response to TG2 bears signs of a primary immune response despite the disease chronicity.

Source continuity

The canonical evidence above remains the authoritative retrieval target.